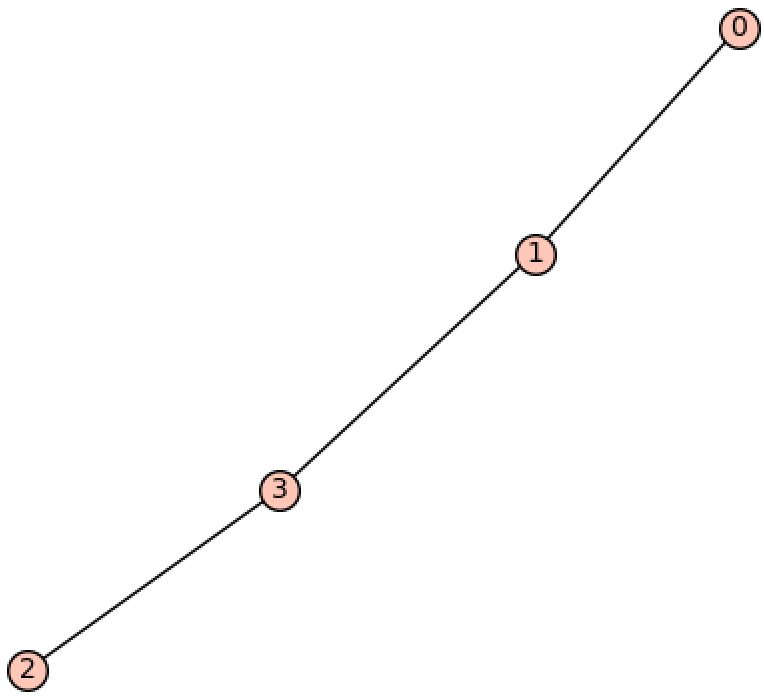
Kernel: SageMath 10.7

In [11]:
```
T = graphs.RandomTree(4)
T.show()
D = T.distance_matrix()
print(D)
print("The determinant of the distance matrix D is :",det(D))
```

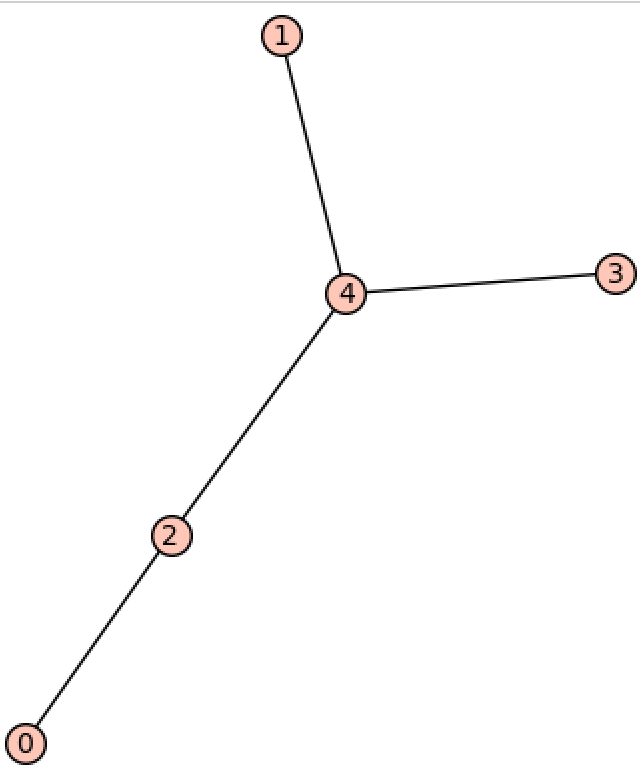
Out[11]:




```
[0 1 3 2]
[1 0 2 1]
[3 2 0 1]
[2 1 1 0]
The determinant of the distance matrix D is : -12
```

In [10]:
```
T = graphs.RandomTree(5)
T.show()
D = T.distance_matrix()
print(D)
print("The determinant of the distance matrix D is :",det(D))
```

Out[10]:

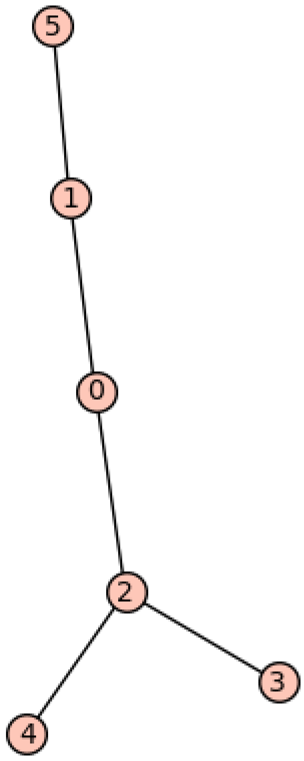



```
[0 3 1 3 2]
[3 0 2 2 1]
[1 2 0 2 1]
[3 2 2 0 1]
[2 1 1 1 0]
The determinant of the distance matrix D is : 32
```

```
T = graphs.RandomTree(6)
T.show()
D = T.distance_matrix()
```

```
print(D)
print("The determinant of the distance matrix D is :",det(D))
```

Out[5]:




```
[0 1 1 2 2 2]
[1 0 2 3 3 1]
[1 2 0 1 1 3]
[2 3 1 0 2 4]
[2 3 1 2 0 4]
[2 1 3 4 4 0]
The determinant of the distance matrix D is : -80
```

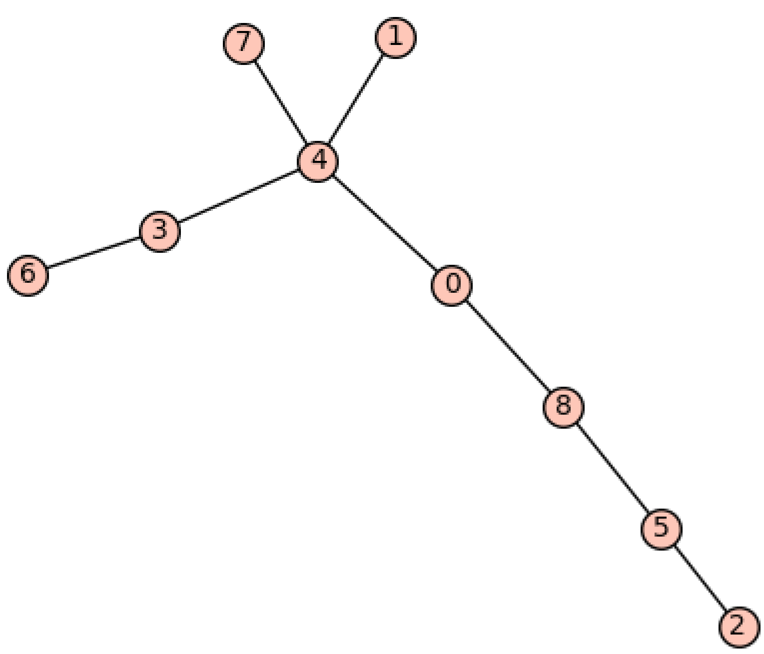
User input Trees and its Distance Matrix

In [12]:

```
n= int(input("Enter the number of vertices"))
T = graphs.RandomTree(n)
T.show()
D = T.distance_matrix()
print(D)
print("The determinant of the distance matrix D is :",det(D))
```

Out[12]:

```
Enter the number of vertices 9
```




```
[0 2 3 2 1 2 3 2 1]
[2 0 5 2 1 4 3 2 3]
[3 5 0 5 4 1 6 5 2]
[2 2 5 0 1 4 1 2 3]
[1 1 4 1 0 3 2 1 2]
[2 4 1 4 3 0 5 4 1]
[3 3 6 1 2 5 0 3 4]
[2 2 5 2 1 4 3 0 3]
[1 3 2 3 2 1 4 3 0]
The determinant of the distance matrix D is : 1024
```